# ON Earth Life

MAGZAZINE

#2

## TOPICS

light - another particle

big thoughts facts

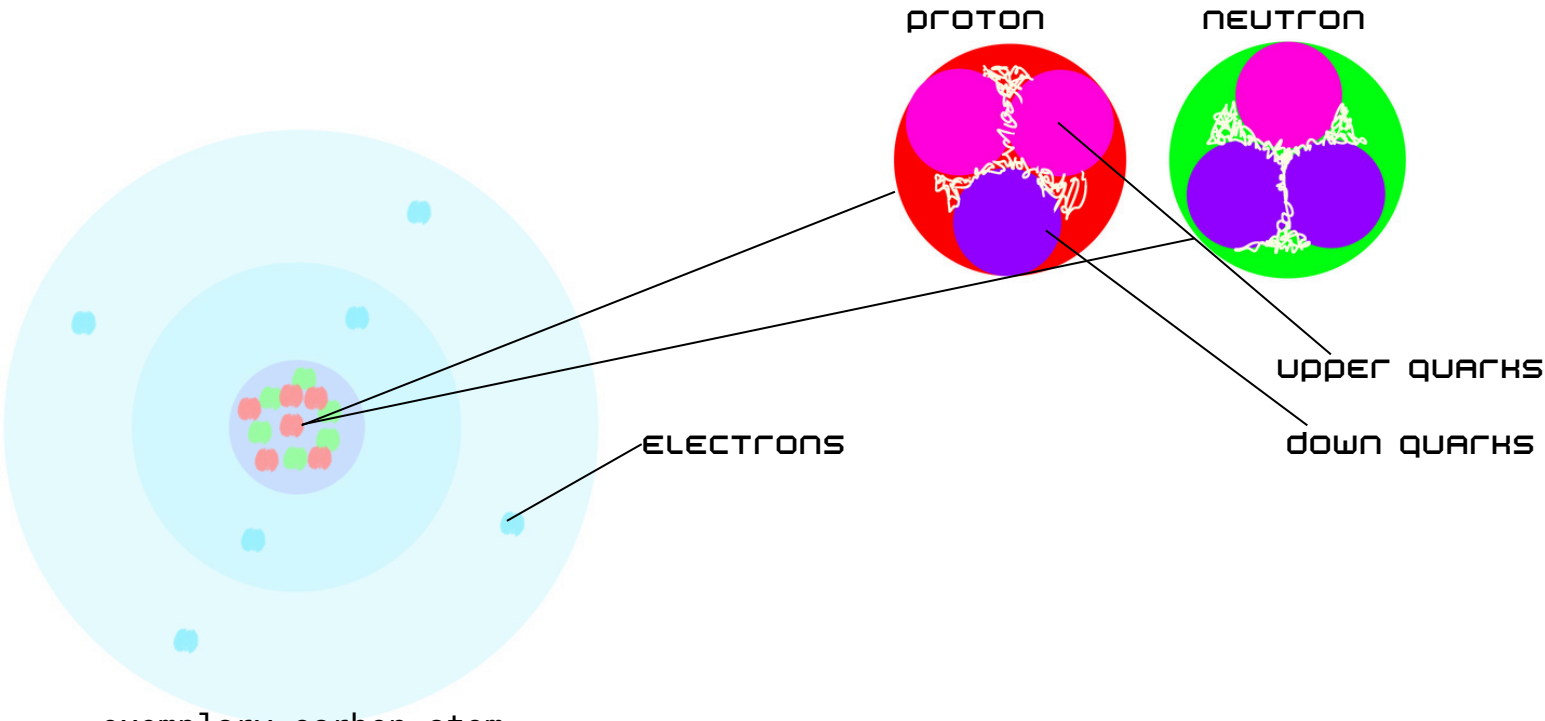
idea for potentially
perfect device

In the previous on Earth Life magazine we spoke about atoms.

quick reminder and sum up on the next page

Atoms are the basic building blocks of our reality - they make up most of the ordinary matter (for instance, your body, a tree or a table).
They typically contain three types of particles: protons, neutrons and electrons (and two of them - protons and neutrons are made of even smaller, particles called quarks).
Every of these particles have energy (as EVERY particle does), and some of these particles have an electric charge (which not every particle has).

Electric charge determines how a particle interacts with the electromagnetic field (explanation in the further part of this magazine). We also know that one part of an atom - the electron - occupies a region around the nucleus (central part made of protons and neutrons), but its position cannot always be assigned a specific place.




exemplary carbon atom

**We also know how many electrons a neutral atom has, as this number equals the number of protons*.**
**in an atoms basic state - as it is described in Mendeleev's periodic table. - We classify atoms into elements based on the number of protons they have.*

**Electrons can be in different specified allowed energies (energy levels), and the number of electrons that can occupy each level is limited.**

**Atoms connect together by sharing electrons.**

**Electrons and quarks unlike protons and neutrons are fundamental particles - not made of any smaller ones**

**tiny fun fact - a Human that weighs 70kg is made of approximately 650 000 000 000 000 000 000 000 000 carbon atoms**

So we know, that crucial part of an atom, which create our reality can be in some way anywhere around the central part, before we observe and capture it ( science can't fully tell what could this mean ).

CONTINUING WITH THE ON REALITY KNOWLEDGE

PHOTONS - LIGHT PARTICLES

**Photons**, like electrons, **are elementary particles.**

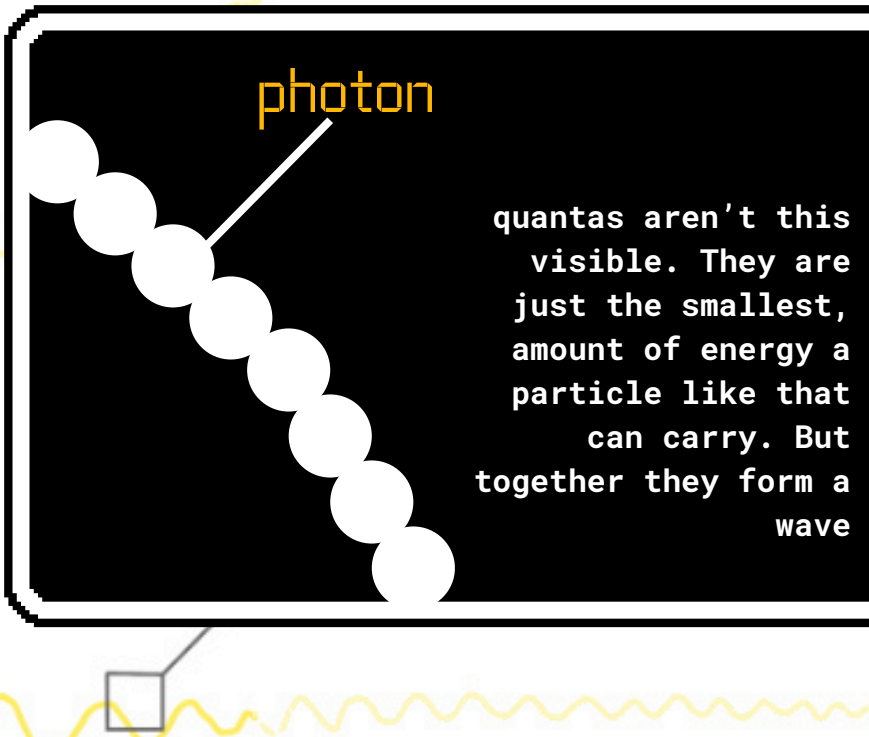
Photons have zero rest mass and no electric charge.

In a vacuum, photons travel at the speed of light.

The speed of light in a vacuum is the maximum speed at which electromagnetic waves propagate through a vacuum.

An electromagnetic wave is a propagating disturbance of the electromagnetic field.

In quantum physics, photons are the quanta of this field.




The electromagnetic field is a field that exists throughout space and time. It can be visible by the activity of electric charges and electric currents, but it also exists and propagates through a vacuum without such 'things'.

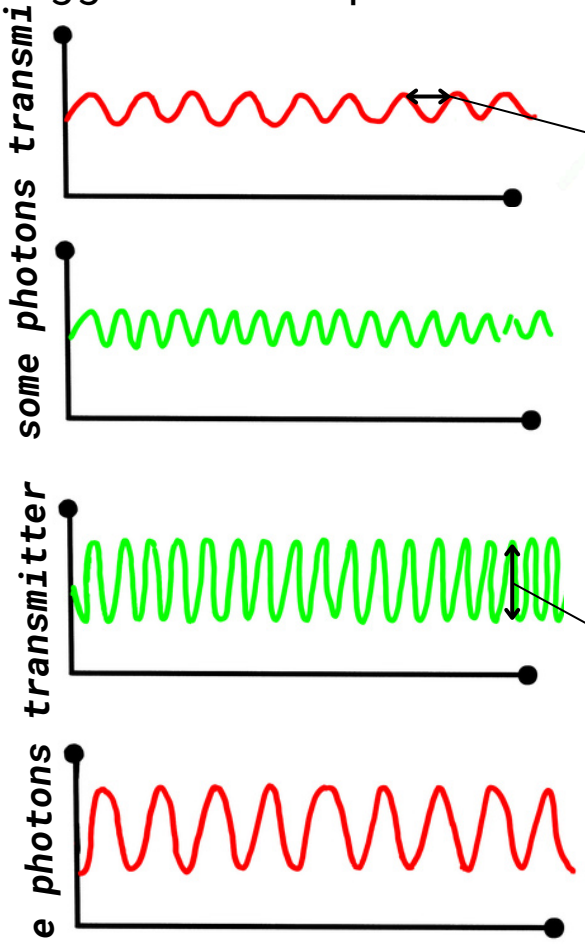
A photon's energy is proportional to the frequency of the electromagnetic radiation it belongs to.

=

A photon's energy determines the frequency of electromagnetic radiation which it is a part of.

radiation = electromagnetic wave

The strength of the emitted radiation is related to the amplitude of the electromagnetic wave, which, in turn, determines the intensity. The stronger the emission the bigger the amplitude




**we percieve light waves with different frequencies as colours:**

**red - smaller frequency (bigger breaks between peaks)**

**green - bigger frequency (smaller breaks between peaks)**

**more vivid/bright green - green frequency in higher amplitude)**

**more vivid/bright red - red frequency in higher amplitude)**

**The amplitude** (which is related to intensity) depends on the strength of the emitted radiation, while the **frequency** (which is related to a colour) is determined by the emission process and the energy of the emitted photons.

The human brain perceives different colours based on signals from photoreceptors in the retina, which respond to different wavelengths.

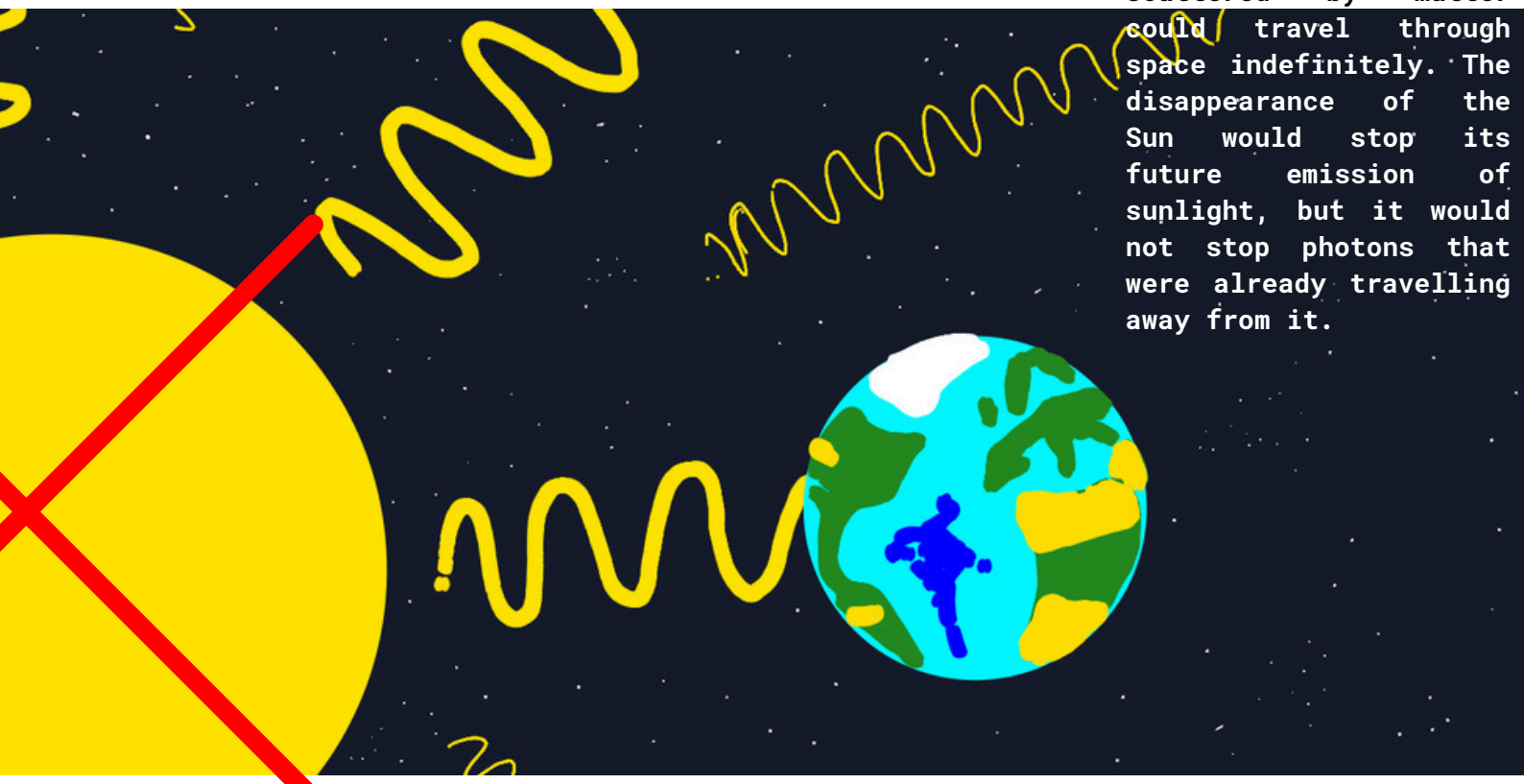
Without photons reaching our eyes, we wouldn't be able to see, because there would be no light for our visual system to detect.

explanation in next pages

Electromagnetic radiation can be emitted by various physical processes, such as changes in the energy states of atoms or the acceleration of electric charges. This radiation carries energy and momentum but has no rest mass.

In a vacuum, electromagnetic radiation can propagate indefinitely unless it interacts with matter or undergoes other physical processes that change its energy or direction.

**If the Sun suddenly disappeared, photons that had already been emitted and were travelling through space would continue the travel. Those that were not absorbed or scattered by matter could travel through space indefinitely. The disappearance of the Sun would stop its future emission of sunlight, but it would not stop photons that were already travelling away from it.**



**Science can describe how wave-like phenomena behave, but why the universe follows the laws that govern their formation and symmetry in this particular wave way remains a deep question.**

SOME

# FUN FACTS

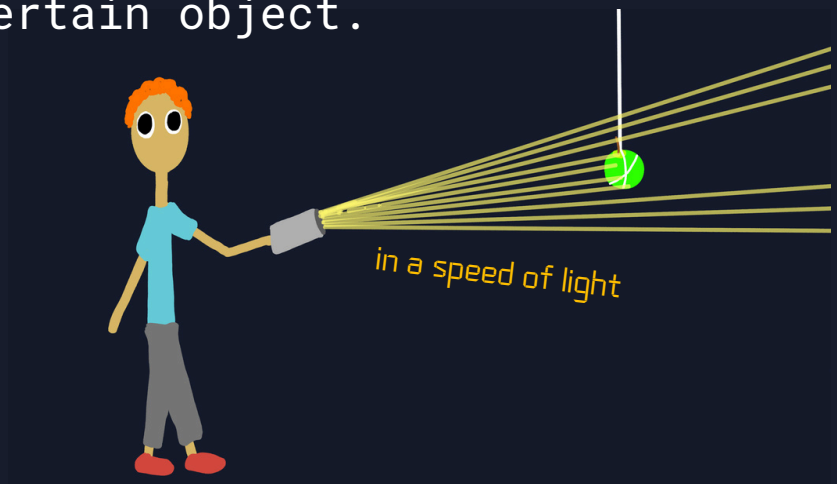
**Why can't we be entirely sure whether what we see is anywhere close to the actual "truth" around us.**

**Mainly because of the structure of our brains.**

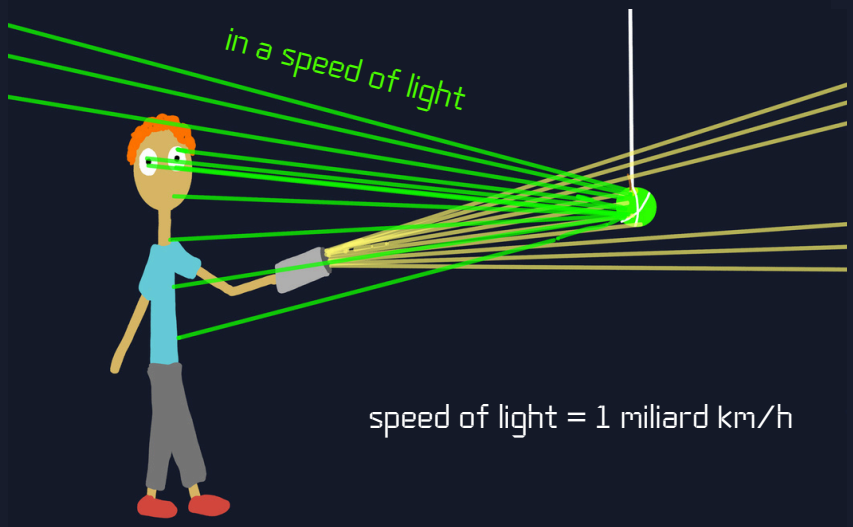
Sight works according to this scheme:

1. Light shines on a certain object.

2. The object absorbs some amount of the light energy and some of it reflects back.

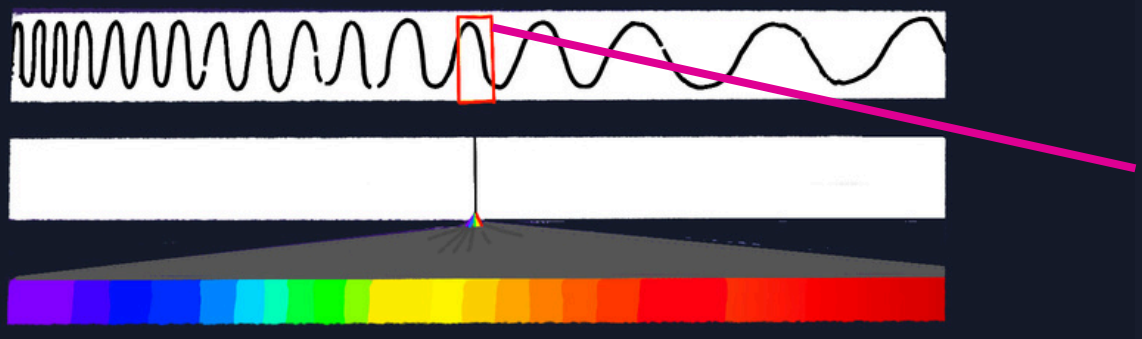



3. Some of the wavelengths of light (there are millions of possible wavelengths) reflected from the object enter our eyes.

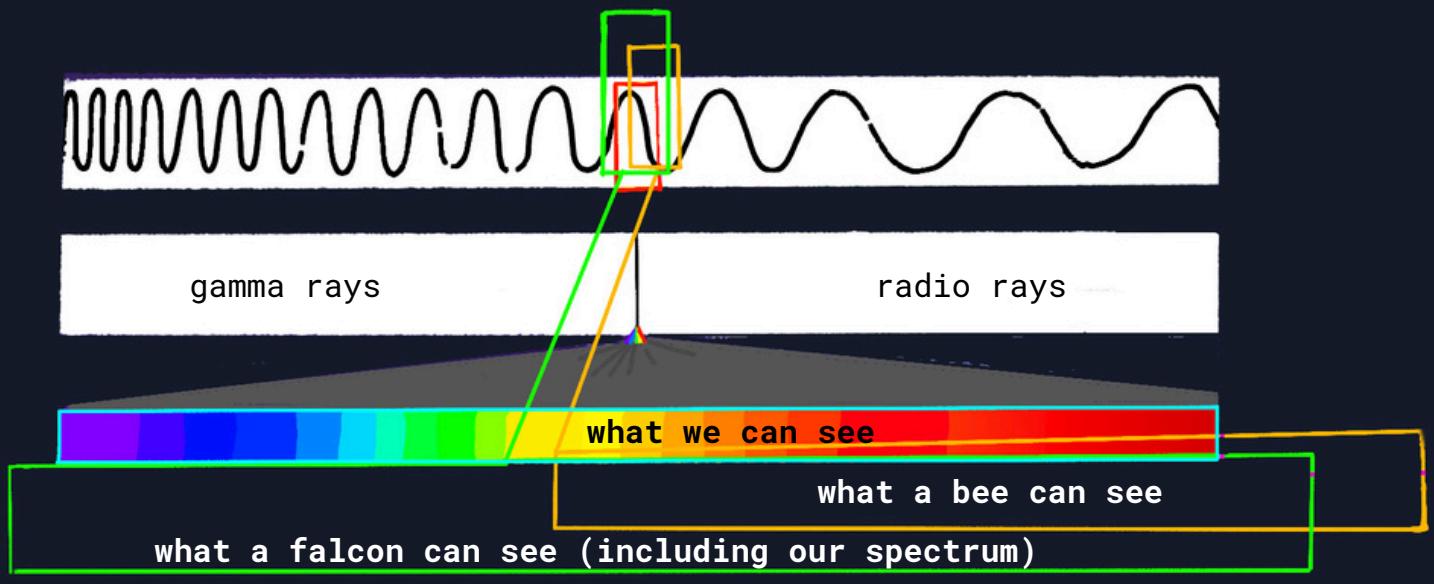



4. They reach photoreceptors in the retina (structures inside our eyes). These photoreceptors convert light into electrical signals, which are then processed by our brain and interpreted as an image you 'see'.

Our human visual system allows us to perceive only a limited spectrum of light ( wavelengths ).
We know how long or short a light wave can be, and which part of this spectrum we can see.



**The electromagnetic spectrum covers an enormous range of frequencies ( wavelengths ), from extremely low-frequency radio waves to extremely high-frequency gamma rays.**




The entire spectrum of light, from what we know is approximately :

**1 Hz ( and less ) - 100 000 000 000 000 000 000 000 Hz**

Our visual system allows us to see approximately this range:

400,000,000,000,000 Hz – 790,000,000,000,000 Hz.

A bee's visual system allows it to detect light at frequencies that extend into the ultraviolet range, which is invisible to us. Approximately:

**460 000 000 000 000 Hz – 1 000 000 000 000 000 Hz**

A falcon's visual abilities also differ from ours, Approximately:

**375 000 000 000 000 Hz – 857 000 000 000 000 Hz**

This most likely means that these animals experience aspects of the world that we cannot experience.

For example, bees can detect ultraviolet patterns on flowers that are invisible to humans.

Some birds, for instance, can use Earth’s magnetic field to help them navigate. This ability is called magnetoreception. Cryptochromes in birds' eyes allows them to somewhat percieve electromagnetic field and that helps them find their way.

However, we currently don’t have a way to experience the world exactly as another animal does, through its own senses and brain.

Therefore, we cannot assume that phenomena we cannot see are non existing.

**And when we consider the small part of the electromagnetic spectrum that we can see compared with the enormous range that remains invisible to our eyes, we can certainly conclude that there is much more to the physical world than we experience directly - and some of it may be difficult for us even to imagine.**

# How can we look into the past

Light travels at the speed of light – around 1 billion km per hour.

It's a speed like any other (even though it's the highest possible speed, as far as we know).
And a light wave is a vehicle like any other (even though it doesn't have any mass).

Therefore, light gets from one point in space to another in some amount of time.

And the enormous speed of light makes our reality seem present, even though what we see is delayed. But with the enormous distances in the Universe, we can clearly notice this delay.

The first example: **light from the Sun**

**the Sun is 149 million kilometers away from Earth.**

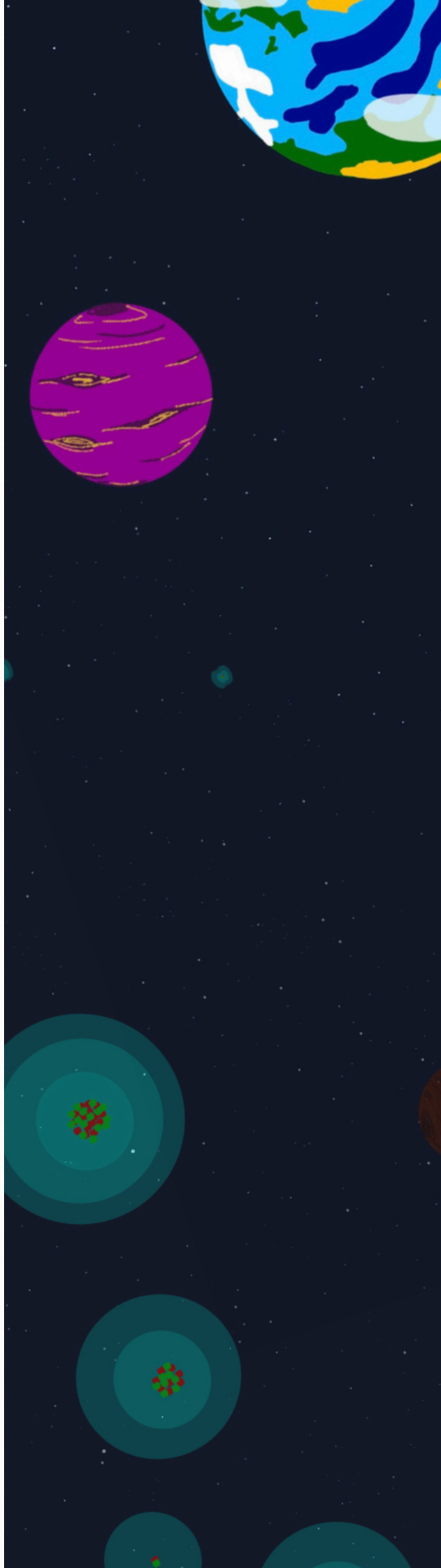
**Its rays travel to us at the speed of light – around 1 billion kilometers per hour.**

**Therefore, the light emitted by the Sun reaches us in around 8 minutes.**

One light-year is a unit of distance corresponding to the distance traveled by light in one Earth year.
 = which is around 9.47 trillion kilometers.

It may seem like a lot; however, there are some objects we observe in the Universe that are located around 100 million light-years away from us.

So, if an object is located 947,000,000,000,000 km away from us, and the light it emits travels at a speed of 1,080,000,000 km/km, It will reach us in around 100 million years.
Therefore, right now we see a light that was emitted 100 million years ago, and the light being emitted right now will reach us in 100 million years as well.
If the star emitting it died, we would know about it only when the last light it emitted reached us.
So, if a star disappeared from our view, it could mean that it actually disappeared 100 million years ago (if the star is 100 million light-years away from us) and that the last light it emitted has only just reached us.



Another example: if someone were 800 light-years away from Earth and had a super-powerful telescope, they would observe Earth through the light that shows what Earth looked like 800 years ago (in the year 1226). The light showing what is happening at this very moment on Earth will reach the observer in 800 years.

**What does this mean?**

**Can we decode waves from far away into a continuous sequence of images?**

**And aren't we already doing that?**

**Therefore, does a photons waves carry images?**

For now, we can only guess.

'ultimate 21st-century useful device for human beings'

Technology on Earth is in constant development.
However, similarly to the best solution for distributing water, we have probably already reached the peak of widely used technology ( phones ) some time ago ( an article on how we managed water much better in the past is available in short form on www.oelfvmag.com ).

side view of a nowa days phone



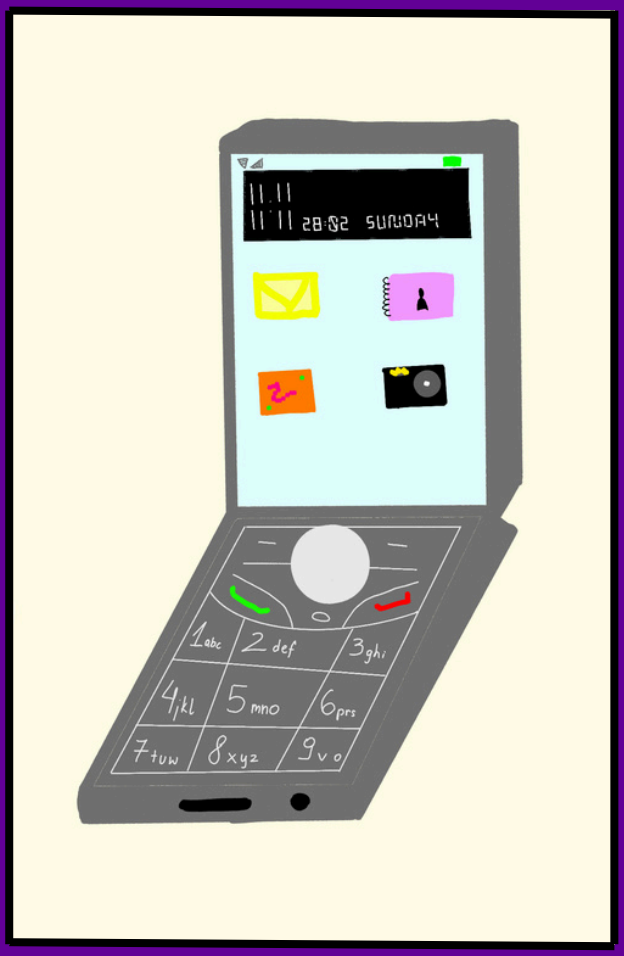

Nowadays, we have flat devices made of easily breakable materials, which most of the time lack storage space and discharge quickly.
However, it wasn't always like that.
We can take past developments and combine them with the knowledge we have today in order to create the 'ultimate 21st-century useful device for human beings'.

Beginning with inspiration from the past, provided by:

- Palm tops / PDAs ( which not many people have heard of )
- Clamshell phones / flip phones ( which many people love )

Clamshells were iconic. However, they weren't extremely functional. They didn't have the ability to download certain apps, didn't have much storage space, and didn't offer the possibility of expanding it. They also had very poor-quality cameras and didn't have touchscreens.

From them, we can mainly take inspiration from the **flip design**.




In order to understand which functions we can use from the palmtop, it may be useful to understand what a palmtop really was.

## Palmtop – PDA – personal digital assistant.

The reason not many people have heard of it is that it was almost completely superseded by smartphones.

It didn't have the ability to make calls.




A palmtop was originally supposed to be a pocket-sized computer. And so, it had most of the functions that a computer has:

- The ability to connect via Wi-Fi / Bluetooth (by the way.. bluetooth are also electromgnetic field waves, in a frequency we cannot percieve).
- A keyboard and all the functions that come with it.
- and many more

Palmtops have a horizontal structure The horizontal design is probably better than vertical one, because it basically allows you to see more. Additionally, the horizontal keyboard has an overall better design because it makes the device look like a game controller ( in this case, it helps you move through the 'game' that is real life )

One of the palmtop's crucial functions is compatibility with compact memory cards.

Thanks to this, there is almost no chance that a PDA will ever run out of storage space if the cards are managed properly.



You can download different files onto different memory cards and use them whenever you need.
Different files – different photo albums, movies, music, documents, books, games, apps, etc.

*In the 'perfected world', social media and games would also be accessible via memory cards. This could help fight the mindless scrolling issue, since opening an app would usually take more time.

For instance, you could use five memory cards at the same time.

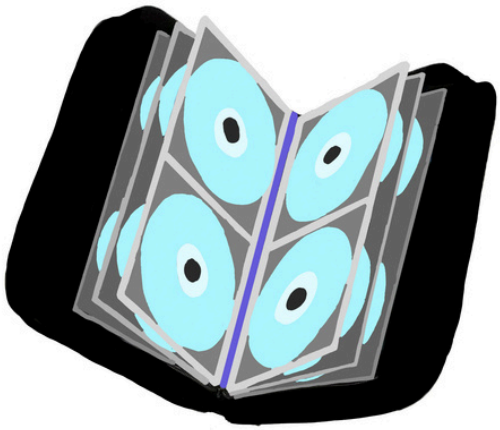
# potential new device, with potentially new software




The solution for storing the cards could be inspired by the CD album era.

But with the albums being much smaller – the size of a memory card, not a CD.

And keeping them together as a chain directly next to the PDA, or arranging them however you prefer.



Now, for ideas that are probably possible with today's knowledge:

- The back of the device made of solar panels.
- An antenna that can connect to a dedicated safety satellite.

Thanks to these functions, the user's overall safety during their experience on Earth could be significantly improved:

Solar panels could provide power to the device, potentially allowing it to **operate almost continuously.**

The antenna could connect to a satellite more effectively and therefore:
Send location information to the user ( which could be displayed on a map built into the new device's software and working even without an internet connection, with major streets, roads, rivers, etc. highlighted ).
Send the user's location and information about the need for help to the appropriate authorities, if necessary.
Imagine this:
There is a satellite you can 'call'.
1 call means medical help.
2 calls mean the police are needed.
3 calls mean the fire department is needed.
4 calls mean helicopter assistance is needed.
It sends your information ( location and type of help needed ) and connects you with the appropriate authority.
Whether you provide more details or not, it will send the right help to you.

Thanks to the antenna and solar-panel back, it could be very helpful in situations such as getting lost somewhere.
And for people who use it for the wrong reasons, there would be a huge fee waiting.

Additional inspiration comes from the materials themselves – both palmtops and clamshells were made from materials much more durable than those used in today's phones.
So we could use that advantage while still keeping the touchscreen function.
It could potentially be a long-lasting device.
Definitely longer-lasting than devices with screens made of fragile materials are.

So,

**From a clamshell, we could use:**

- The ability to call other devices.
- The presence of front and back cameras.
- The flip design.
- Every function associated with a phone.

**From a palmtop, we could use:**

- The horizontal design.
- Memory card slots.
- Every function associated with a computer.

**From today's knowledge, we could use:**

- Solar panel back.
- A durable touchscreen.
- Good-quality cameras.
- Emergency anthena

A device that can have almost unlimited and, well, visually organised storage.

A device that can work without access to electricity.

A device that can locate you and search for potentially needed help wherever you are.

A device that works both as a phone and a computer and has the shape of a video game controller.



And the design can, of course, be as you want it to be.

The company that changed the trajectory of PDA development was Apple. They made their version of a PDA without a keyboard… From that moment on, modern smartphone design began to take over.

And the same company could:

**reverse this trend and introduce the world to more durable device with different shape. Especially since there is an ongoing trend of returning to old designs and truly fun-looking technology.**

Extra ideas: Solar-panel bricks



In some countries, making very durable bricks from processed plastic waste is gaining on popularity.

for istance: Gjenge makers in Kenia

It could be a promising way to manage the huge amounts of plastic waste appearing on the planet.

Similarly, there may be a chance to do the same with old solar panels, since they are difficult to recycle and take a very long time to break down.

Then perhaps appropriate processing – shredding, mixing with the right additives, and pressing – could give the panels a second life?

If not you, then maybe someone else.
Feel free to share the magazine.
There has to be a soul that will move masses with the right vision.

on Earth Life



peace, Aeria Tellar
from Venus



